# 2.2. Electric Flux & Electric Potential

**Why do birds sitting on high voltage power lines do not get electrocuted?**
`es03 with hat`

- need to understand:
    - concept of electric potential and voltage
    - relation to electric potential energy
    - deeper understanding of electric fields, i.e. electric flux & Gauss's law
- start by revisiting electric field experimentally `es05`

## Electric flux for uniform electric field

`sim: rectangle in field`

- For a uniform electric field, the electric flux is defined as

$$\Phi_E = \vec{\mathbf{E}}\vec{\mathbf{A}}$$

  where $\vec{\mathbf{A}}$ is the vector perpendicular to the surface with magnitude $A$
- Using $\cos\theta$ as the angle between $\vec{\mathbf{E}}$ and $\vec{\mathbf{A}}$, the equation can be rewritten as

$$\Phi_E = EA\cos\theta = E_\perp A = EA_\perp$$

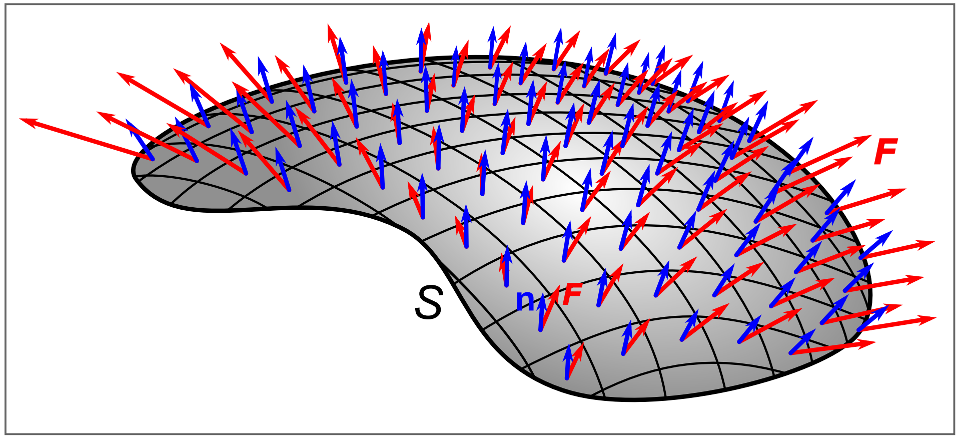
- The number of field lines $N$ passing through an area perpendicular to the field is proportional to the electric flux

$$N \propto E_{\perp} A = \Phi_E$$

# Electric flux for non-uniform electric field

- arbitrary surface can be decomposed into infinitesimal areas $d\vec{\mathbf{A}}$
- for each $d\vec{\mathbf{A}}$, associated field is uniform
- integral over surface gives the total flux:

$$\Phi_E = \int \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}}$$



# Gauss’s law

- named after Karl Friedrich Gauss (1777-1855)
- **Gauss's law**: **Electric flux through a closed surface is equal to the net enclosed charge divided by the permittivity of free space**:

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{enc}}{\epsilon_0}$$

*Note: You find different notations of the integral such as* $\int_S$ *or* $\iint_S$

- **by conventions**
  - $d\vec{\mathbf{A}}$ points outwards from the surface of the enclosed volume
  - flux leaving the surface is positive
  - flux entering the surface is negative
- **consequences on net flux**:
  - if $\Phi_E$ is positive, there is a net flux out of the volume
  - if $\Phi_E$ is negative, there is a net flux into the volume

- if $\Phi_E = 0$, there is no net flux

## Gauss's law (cont')

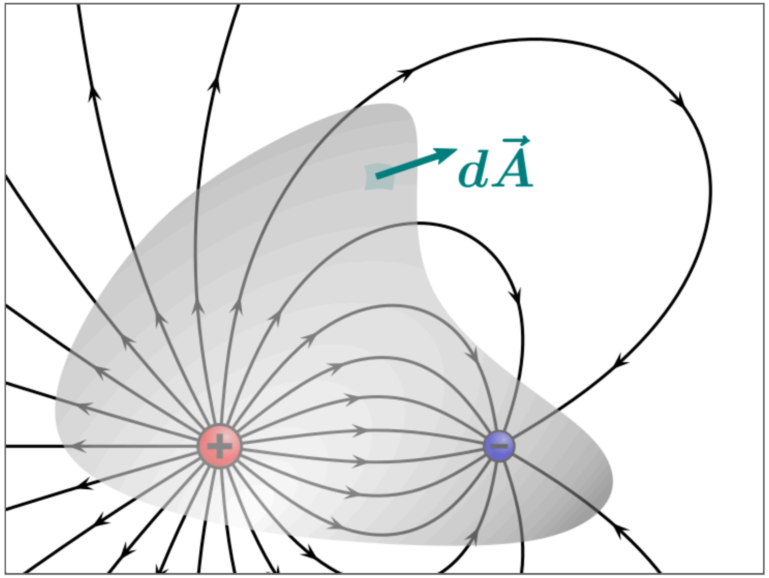
`sim: circle in field`

- flux through surface is **independent** of:
  - distribution of the enclosed charge within the volume
  - charges outside the surface that may affect the position but not the number of field lines
- **principle of superposition applicable**:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \oint (\sum_i \vec{\mathbf{E}}_i) \cdot d\vec{\mathbf{A}} = \sum_i \frac{Q_i}{\epsilon_0} = \frac{Q_{enc}}{\epsilon_0}$$

# Examples for Gauss's law

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{enc}}{\epsilon_0} > 0$$

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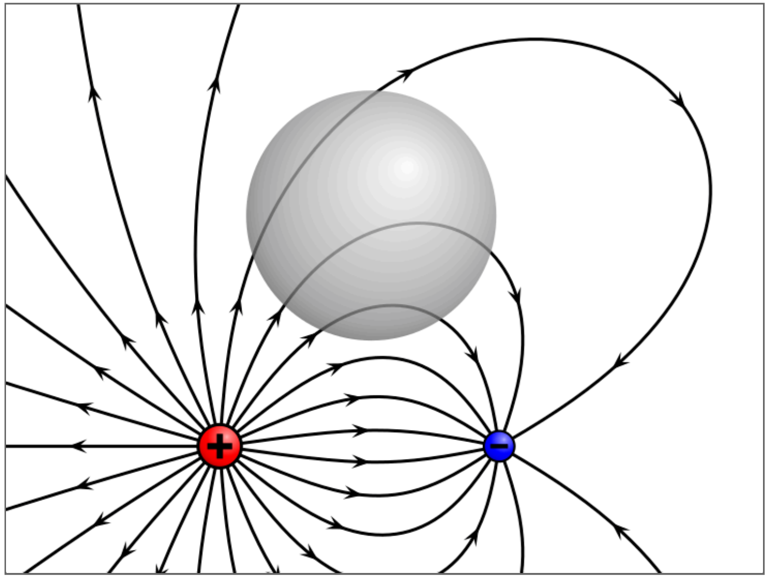
# Examples for Gauss's law (cont')

$$\Phi_E = \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{enc}}{\epsilon_0} = 0$$



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# Gauss's law: Relation to Coulomb's law

`thought experiment`

- consider a spherical surface (radius $r$) with a single enclosed point charge $Q_{enc}$ at the center
- the $\vec{\mathbf{E}}$ field is oriented radially with the same magnitude everywhere on the surface
- field lines penetrate the spherical surface perpendicular to $d\vec{\mathbf{A}}$
- surface area of the sphere is $\oint dA = 4\pi r^2$
- This leads to Coulomb's law in electric field form:

$$\frac{Q_{enc}}{\epsilon_0} = \oint \vec{\mathbf{E}} d\vec{\mathbf{A}} = E \oint dA = 4\pi r^2 E$$

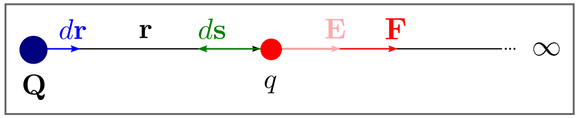
$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} \quad \text{with} \quad Q_{enc} = Q$$

# Complementary nature of Coulomb's and Gauss's law

| | Coulomb's Law | Gauss's Law |
|---|---|---|
| **Equation** | $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} \leftrightarrow F = \frac{1}{4\pi\epsilon_0}\frac{Q_1Q_2}{r^2}$ | $\oint \vec{\mathbf{E}}\, d\vec{\mathbf{A}} = \frac{Q_{enc}}{\epsilon_0}$ |
| **Focus** | Field/force & (point) charges | Flux through surface & enclosed net charge |
| **Applicability** | Point charges, any distribution (complex) | Mostly used for symmetrical distributions, although applicable to any case |
| **Relevant Charge Sources** | All charges present contribute | Only enclosed charges contribute to flux calculation |

# Electricity & energy?

`es01 - elest. Bell`

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## Electric potential energy

- concept of **conservation of energy** in electricity is analogous to mechanics
- **electric potential energy is defined for conservative forces**, e.g. the electrostatic force $\vec{\mathbf{F}} = \vec{\mathbf{E}}q$ (conservative force $\rightarrow$ path independence)
- **in uniform electric field**, the **work** to move a test charge $q$ over a distance $d$ is:

$$W = Fd = qEd$$

- change in **potential energy** $\Delta U$ from point $A$ to $B$ is:

$$\Delta U = U_B - U_A = -W_{AB} = -q\,E\,d_{AB}$$

- note the similarity to mechanics, i.e. $mgh$, but as we have two types of charge:
    - like charges $\rightarrow$ high potential energy when close

- opposite charges $\rightarrow$ low potential energy when close

# Voltage: The difference in electric potential

`es03 w\ tube`

- **electric potential** $V$ is defined as the **electric potential energy per unit charge**

$$V = \frac{U}{q}$$

- **voltage is the difference in potential** between points $A$ and $B$:

$$V_{BA} = V_B - V_A \quad \text{in } [\text{J}/\text{C}] = [\text{V}]$$

- acknowledging Alessandro Volta, the unit is called **volt** [V]
- **reference point** (typically ground or infinity) is chosen where $V = 0$

# Gravitational analogue

In gravity near Earth, potential energy: $U = mgh$
Divide by mass $m$, to get **gravitational potential**:

$$V_g = \frac{U}{m} = gh \quad \leftrightarrow \quad V = \frac{U}{q}$$

**Intuition**

- Height $h$ ↔ electric potential $V$
- Mass $m$ ↔ charge $q$
- $\rightarrow$ heavy object ↔ large charge
- $\rightarrow$ same “height” → same potential, regardless of mass/charge

## Putting it all together

- **electric potential** is defined at a **single point** in space
- only potential **differences**, i.e. voltage, are **measurable**
- **reference point** with zero potential is **defined arbitrarily**
- combining the definitions gives

$$V_{BA} = V_B - V_A = \frac{U_B}{q} - \frac{U_A}{q} = \frac{\Delta U_{BA}}{q} = -\frac{W_{BA}}{q}$$

- **work** performed on a charge is given by

$$W_{BA} = Fd = -qV_{BA}$$

- $\rightarrow$ **Voltage (potential difference) tells you how much energy per unit charge is available between two points.**

## The electron volt

- electron volt (eV) is the energy gained by an electron when moving through a potential difference of 1 V

$$1\,\mathrm{eV} \approx 1.60 \times 10^{-19}\,\mathrm{J}$$

- handy when dealing with small energies, i.e. charged particles

## Relating electric potential & electric field

- analogously to mechanics, change in potential energy $\Delta U_{BA}$ from $A$ to be $B$ along the path $\vec{\mathbf{l}}$ is defined as:

$$\Delta U_{BA} = -W = -\int_A^B \vec{\mathbf{F}} d\vec{\mathbf{l}}$$

- we know $V = \frac{U}{q}$ and $\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q} \leftrightarrow \vec{\mathbf{F}} = \vec{\mathbf{E}}q$, therefore:

$$V_{BA} = -\frac{1}{q} q \int_A^B \vec{\mathbf{E}} d\vec{\mathbf{l}}$$

$$V_{BA} = -\int_A^B \vec{\mathbf{E}} d\vec{\mathbf{l}}$$

# Relating electric potential & electric field (cont'd)

$$V_{BA} = -\int_A^B \vec{\mathbf{E}} d\vec{\mathbf{l}}$$

- in **uniform electric field**, the integral simplifies to

$$V_{BA} = -Ed \quad \text{with distance}\, d$$

- for an infinitesimal change, the relation is

$$dV = -\vec{\mathbf{E}}\, d\vec{\mathbf{l}}$$

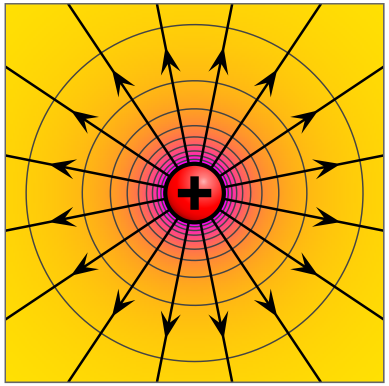
- therefore, electric field is the gradient of the electric potential:

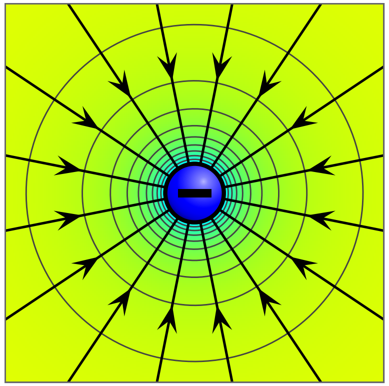
$$\vec{\mathbf{E}} = -\mathrm{grad} V = -\nabla V$$

- $\Rightarrow$ **electric potential decreases in the direction of the electric field**

# Electric potential of a point source

- Start from voltage definition: $V_{BA} = -\int_A^B \vec{E} \cdot d\vec{l}$
- Point charge field: $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$
- Radial symmetry ⇒ $\vec{E} \cdot d\vec{l} = E\,dr$
- Integrate: $V_{BA} = -\int_{r_A}^{r_B} \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}\,dr = -\frac{Q}{4\pi\epsilon_0}\left(-\frac{1}{r_B} - (-\frac{1}{r_A})\right) = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$
- Choose reference: $V(\infty) = 0$
- Coulomb potential: $V(r) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$





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# Equipotential lines & surfaces

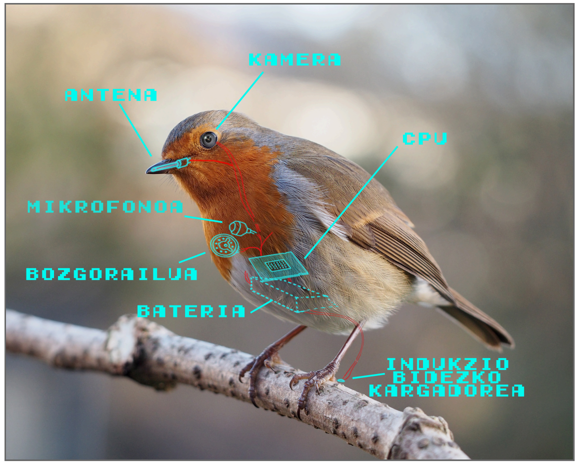
`sim: equipotential`

- equipotential lines (2D) and surfaces (3D) represent regions where the **electric potential is constant**
- **moving perpendicular** to the electric field ($\vec{\mathbf{E}}_{\perp} d\vec{\mathbf{l}}$) does not change the potential

# Revisiting initial question

**Why do birds sitting on high voltage power lines not get electrocuted?**

- Feet nearly same potential, negligible voltage difference
- Current prefers low-resistance wire path, not high resistance path through bird
- Virtually no current $\rightarrow$ no electric shock
- Bridging potentials causes dangerous current flow



# Revisiting initial question: Back-of-the-envelope calculation

Assume:

- Line current: $I \approx 500\,\mathrm{A}$
- Line resistance: $R' \approx 0.05\,\Omega/\mathrm{km}$
- Distance between feet: $d \approx 5\,\mathrm{cm}$
- Bird resistance: $R_{\mathrm{bird}} \sim 10^3\,\Omega$

Resistance between the feet:

$$R = R'd = 0.05 \cdot 5 \times 10^{-5} = 2.5 \times 10^{-6}\,\Omega$$

Voltage difference between the feet:

$$\Delta V = IR = 500 \cdot 2.5 \times 10^{-6} = 1.25 \times 10^{-3}\,\mathrm{V} \approx 1.3\,\mathrm{mV}$$

*For comparison, 90 to 140 mV are required to for human heart contractions*

Current through the bird (overestimation as $R_{\mathrm{wire}} \ll R_{\mathrm{bird}}$):

$$I_{\mathrm{bird}} = \frac{\Delta V}{R_{\mathrm{bird}}} = \frac{1.3 \times 10^{-3}}{10^{3}} \approx 1.3 \times 10^{-6}\,\mathrm{A} \approx 1\,\mu\mathrm{A}$$

# Unconventional ways to make light

`es31`